

Practical no:- 12

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Input:-

```
print("LOCATION COORDINATE PROCESSING SYSTEM")
```

```
locations = [
```

```
    ("Tasgon", 18.5204, 73.8567),
```

```
    ("Vita", 19.0760, 72.8777),
```

```
    ("Jath", 19.9975, 73.7898),
```

```
    ("Atapadi", 17.3850, 78.4867)
```

```
]
```

```
print("\n1. Original Locations:")
```

```
print(locations)
```

```
print("\n2. Indexing Tuple Elements:")
```

```
first_location = locations[0]
```

```
print("Complete Tuple:", first_location)
```

```
print("Name:", first_location[0])
```

```
print("Latitude:", first_location[1])
```

```
print("Longitude:", first_location[2])
```

```
print("Last Element:", first_location[-1])
```

```
print("Second Last Element:", first_location[-2])
```

```
print("\n3. Tuple Unpacking:")
```

```
name, latitude, longitude = first_location
```

```
print("Name:", name)
```

```
print("Latitude:", latitude)
```

```
print("Longitude:", longitude)

print("\n4. Tuple Slicing:")

print("Original Tuple:", first_location)

print("First Two Elements:", first_location[:2])

print("Last Two Elements:", first_location[1:])

print("Only Coordinates:", first_location[1:3])

print("Reverse Tuple:", first_location[::-1])

print("\n4. Sorted by Latitude:")

sorted_by_latitude = sorted(

    locations,

    key=lambda location: location[1]

)

for name, latitude, longitude in sorted_by_latitude:

    print(f"{name}: ({latitude}, {longitude})")

print("\n5. Sorted by Longitude:")

sorted_by_longitude = sorted(

    locations,

    key=lambda location: location[2]

)

for name, latitude, longitude in sorted_by_longitude:

    print(f"{name}: ({latitude}, {longitude})")

print("\n6. Tuple Mutability:")

print("Original Tuple:", first_location)

print("Tuples are immutable.")

print("Therefore, tuple elements cannot be changed directly.")
```

Output:-

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Admin\revati> & 'C:\Users\Admin\local\bin\python3.14.exe' 'c:\Users\Admin\vscode\extensions\ms-python.debug
' '51634' '--' 'c:\Users\Admin\revati\practical 12.py'
LOCATION COORDINATE PROCESSING SYSTEM

1. Original Locations:
[('Tasgon', 18.5204, 73.8567), ('Vita', 19.076, 72.8777), ('Jath', 19.9975, 73.7898), ('Atapadi', 17.385, 78.4867)]

2. Indexing Tuple Elements:
Complete Tuple: ('Tasgon', 18.5204, 73.8567)
Name: Tasgon
Latitude: 18.5204
Longitude: 73.8567
Last Element: 73.8567
Second Last Element: 18.5204

3. Tuple Unpacking:
Name: Tasgon
Latitude: 18.5204
Longitude: 73.8567

4. Tuple Slicing:
Original Tuple: ('Tasgon', 18.5204, 73.8567)
First Two Elements: ('Tasgon', 18.5204)
Last Two Elements: (18.5204, 73.8567)
Only Coordinates: (18.5204, 73.8567)
Reverse Tuple: (73.8567, 18.5204, 'Tasgon')

4. Sorted by Latitude:
Atapadi: (17.385, 78.4867)
Tasgon: (18.5204, 73.8567)
Vita: (19.076, 72.8777)
Jath: (19.9975, 73.7898)

5. Sorted by Longitude:
Vita: (19.076, 72.8777)
Jath: (19.9975, 73.7898)
Tasgon: (18.5204, 73.8567)
Atapadi: (17.385, 78.4867)

6. Tuple Mutability:
Original Tuple: ('Tasgon', 18.5204, 73.8567)
Tuples are immutable.
Therefore, tuple elements cannot be changed directly.
PS C:\Users\Admin\revati>
```